

NATURAL LANGUAGE PROCESSING

Foundations, Web Scraping, Sentiment Analysis & Applications

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Foundations and Basic Concepts
- 2 Web Scraping and Data Collection
- 3 Sentiment Analysis
- 4 Applications and Integration



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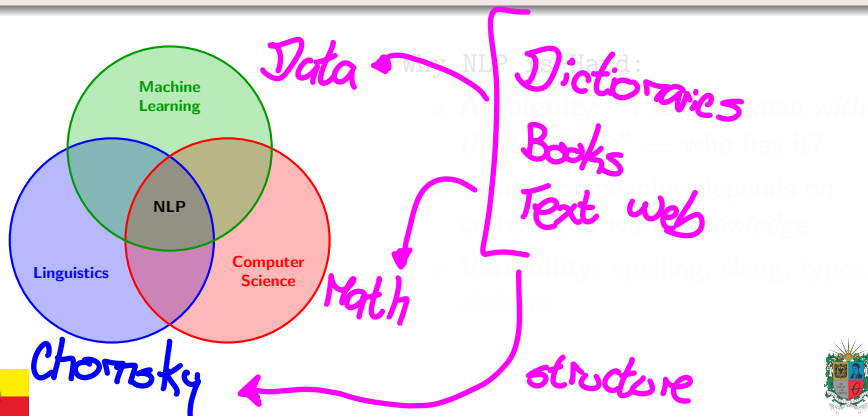


What is Natural Language Processing?

understand / interpretation like on human

Definition

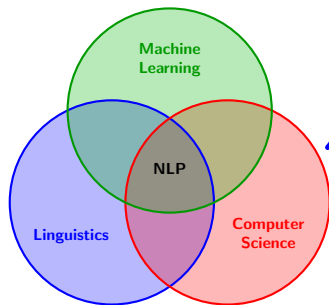
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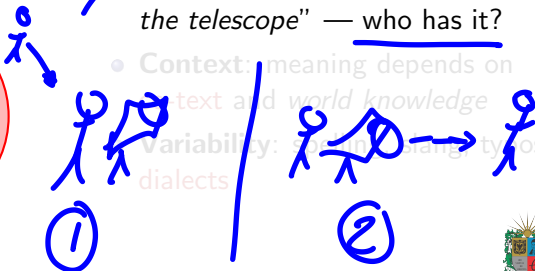


Why NLP is Hard: *tact subject*

• **Ambiguity:** "I saw the man with the telescope" — who has it?

• **Context:** meaning depends on text and world knowledge

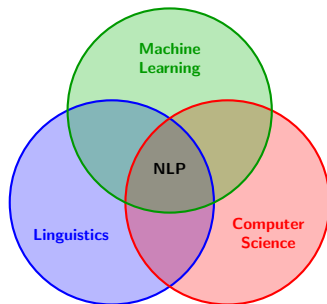
• **Variability:** recognizing typos, dialects



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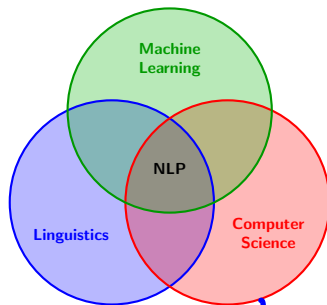
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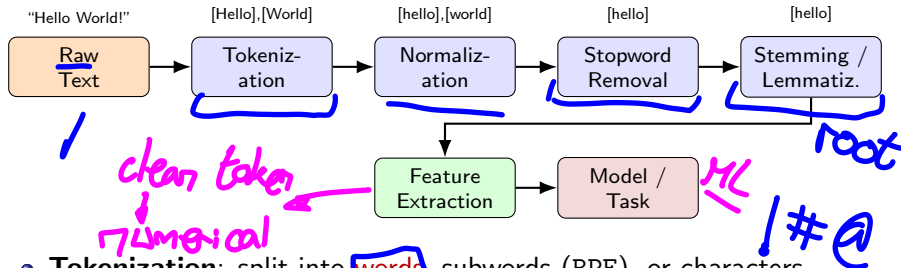
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How a machine can do it?



The NLP Processing Pipeline



- **Tokenization:** split into words subwords (BPE), or characters
- **Normalization:** lowercase, accent removal, punctuation stripping
- **Stemming:** running → run; **Lemmatization:** better → good
(grammatically correct root)

• **stopword:** the, is, at, on
↳ repetition



Text Representation: From Words to Vectors

str $\rightarrow$ [R]

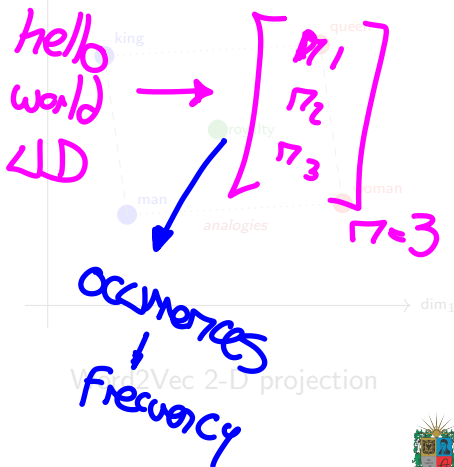
Bag-of-Words & TF-IDF

- Doc $\rightarrow \mathbf{v} \in \mathbb{R}^{|V|}$; order ignored
- $\text{tfidf}(t, d) = \text{tf} \cdot \log \frac{|D|}{\text{df}_t}$
- Downweights common, upweights rare terms

Word Embeddings

- Dense $\mathbb{R}^d$; semantic similarity $\approx$ distance
- king - man + woman $\approx$ queen

Bow



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1) el perro muerde al niño
 log $\rightarrow$ Frequency

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2) helb
 1) $\hookrightarrow$ D $\rightarrow$ B

Reve he $\hookrightarrow$ D $\hookrightarrow$ D he

Word2Vec 2-D projection



Text Representation: From Words to Vectors

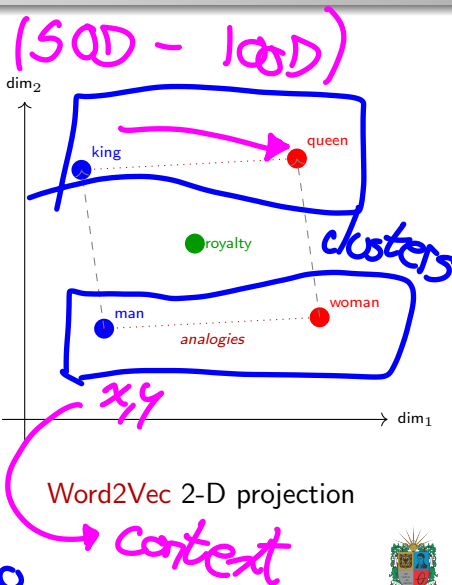
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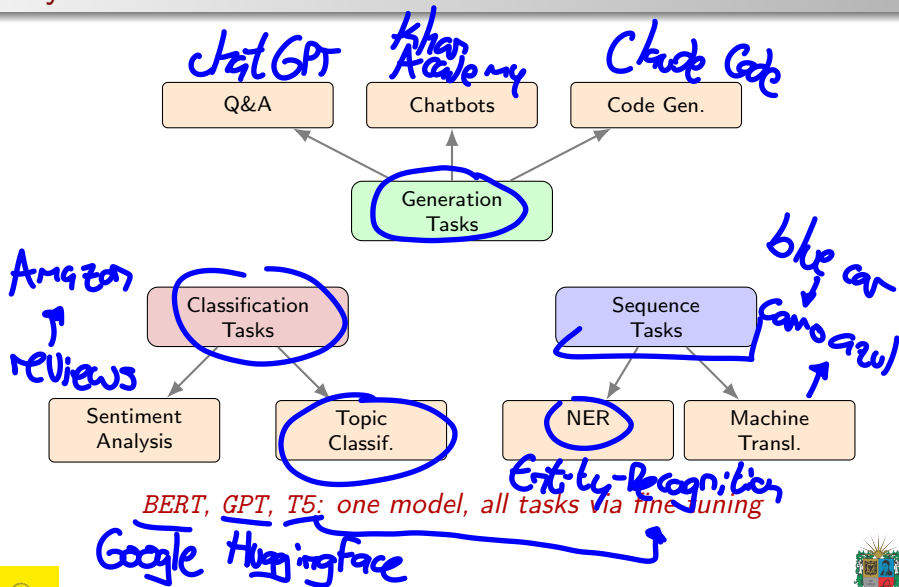
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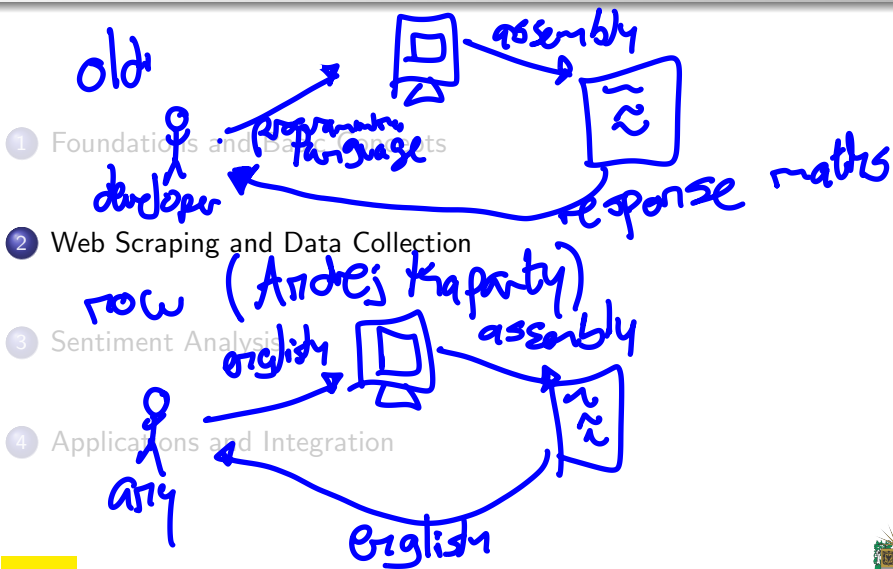
caso $\leftrightarrow$ *edificio*
caso $\leftrightarrow$ *caso*



Key NLP Tasks Overview



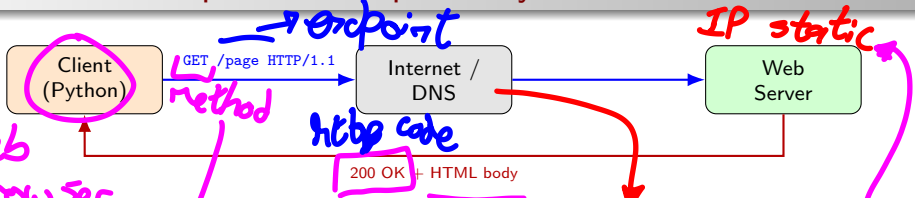
Outline



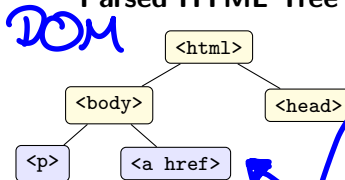
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The HTTP Request — Response Cycle



Parsed HTML Tree



AJAX

google.com - 172.21.17.1
chatgpt - 172.64.135.209

`<html/>`
`<head> </head>`
`<body>`
`<p>...</p>`
`<a href>...</a>`

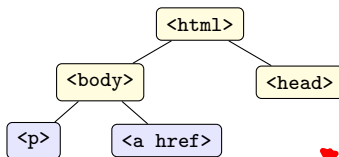
C → POST
R → GET ✓
P → PATCH
D → DELETE



The HTTP Request — Response Cycle



Parsed HTML Tree



- Status: 200 OK | 403 Forbidden | 429 Rate Limited

- Dynamic page: JS renders after HTTP response

→ need Selenium

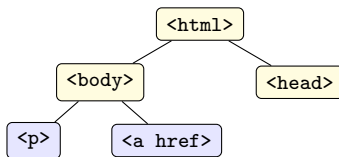
Free quota



The HTTP Request — Response Cycle



Parsed HTML Tree



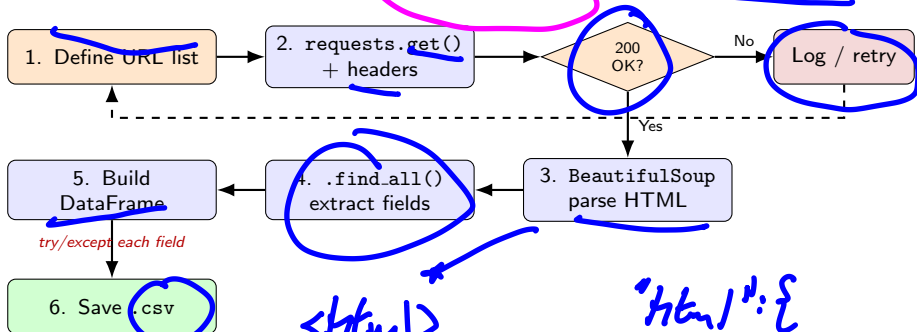
static web pages

- Status: 200 OK | 403 Forbidden | 429 Rate Limited
- Dynamic pages: JS renders *after* HTTP response

need Selenium



Scraping Workflow with requests + BeautifulSoup



- `time.sleep(1--5)` between requests (politeness delay)
- Always respect robots.txt terms of Service

Handwritten HTML tags:

```

<html>
<body>
<div>
<section>
<a>
</a>

```

Handwritten CSS selectors:

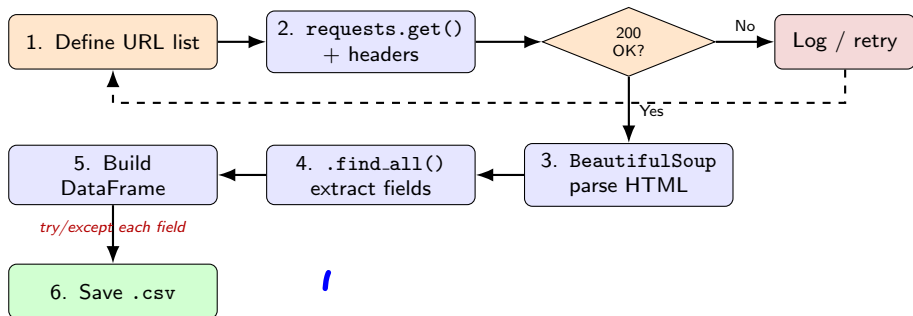
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```



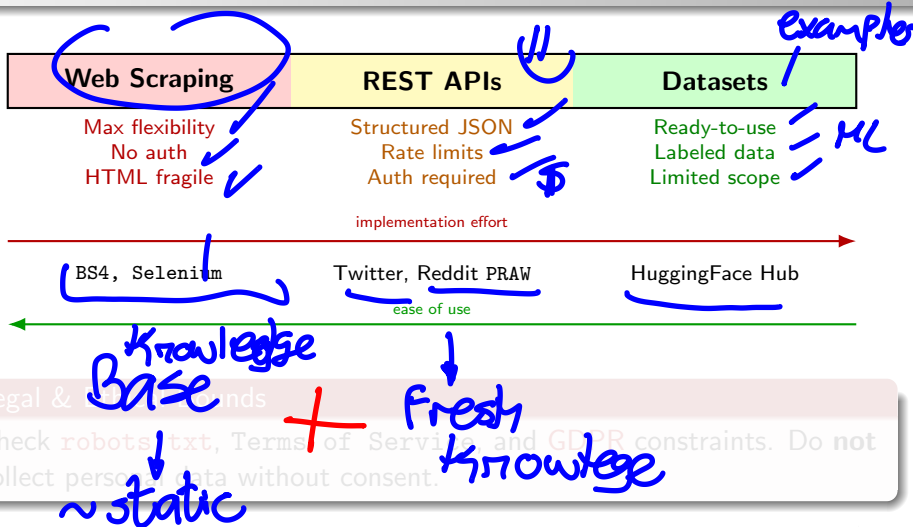
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- Add `time.sleep(1--3)` between requests (politeness delay)
- Always respect `robots.txt` and Terms of Service



Scraping vs. APIs vs. Ready Datasets



Scraping vs. APIs vs. Ready Datasets

Web Scraping	REST APIs	Datasets
Max flexibility No auth HTML fragile	Structured JSON Rate limits Auth required	Ready-to-use Labeled data Limited scope
implementation effort		
BS4, Selenium	Twitter, Reddit PRAW	HuggingFace Hub
ease of use		

Legal & Ethical Bounds

Check robots.txt, Terms of Service, and GDPR constraints. Do **not** collect personal data without consent.

NY Times

~contract



Outline

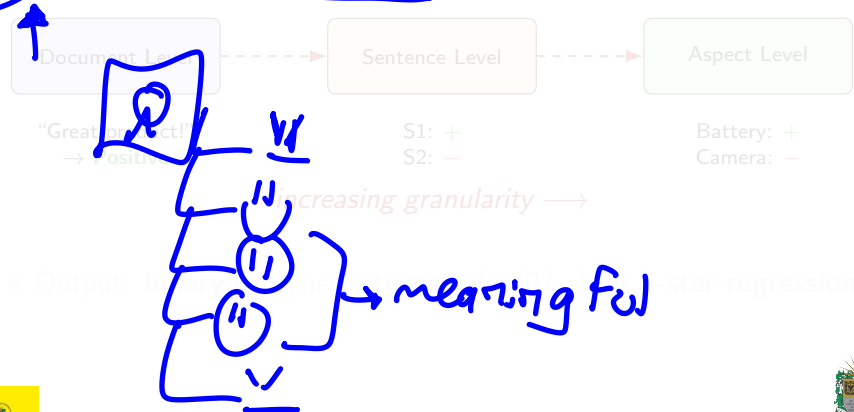
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What is Sentiment Analysis?

Definition

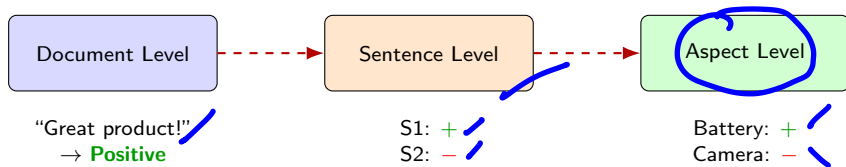
Automatically identify the **emotional tone** of text: positive, negative, neutral — at multiple granularity levels.



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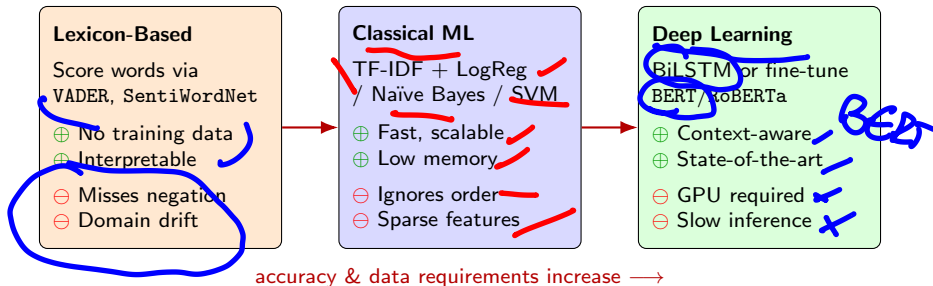


increasing granularity →

Output: **binary** (pos/neg), **ternary** (+/0/-), or **5-star regression**



Three Approaches: Lexicon, ML, Deep Learning



AI → NLP



Evaluation: Metrics and Confusion Matrix

Beyond Accuracy

Accuracy is misleading on imbalanced data. Always report per-class metrics.

classification

Confusion Matrix

high prec.

	Pos (pred)	Neg (pred)
Pos (actual)	TP P → P	FN P → N
Neg (actual)	FP N → P	TN N → N

high rec.

$$\frac{TP + TN}{\text{TOTAL}}$$

$\text{Precision} = \frac{TP}{TP + FP}$
 $\text{Recall} = \frac{TP}{TP + FN}$
 $F1 = 2 \cdot \frac{P \cdot R}{P + R}$

tweet	act. sentiment	pred. anal. sent
x_1	P	P
x_2	N	P
x_3	P	N



Evaluation: *Metrics and Confusion Matrix*

Beyond Accuracy

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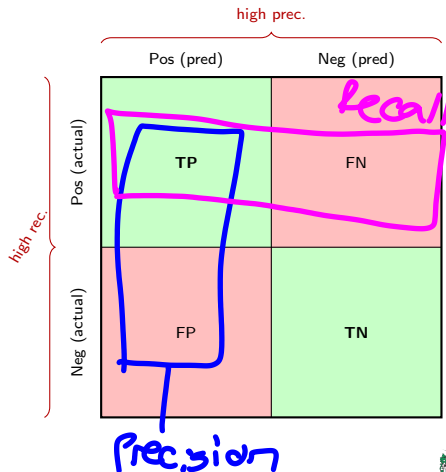
- Precision = $\frac{TP}{TP + FP}$

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- $F1 \approx \frac{2 \cdot P \cdot R}{P + R}$

- F1-macro: average over all classes

Confusion Matrix



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F_{BETA}

Confusion Matrix

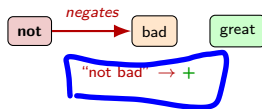
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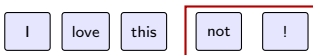
Sentiment Analysis: *Linguistic Challenges*

- **Negation:** "Not bad" → positive
- **Sarcasm:** "Great, another Monday"
- **Domain shift:** medical $\neq$ movie vocabulary
- **Multilingual:** cultural conventions differ
- **Subjectivity:** "Film runs 120 min" has no sentiment

Negation Scope



Context Window



window= 3

Practical Tip

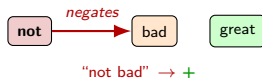
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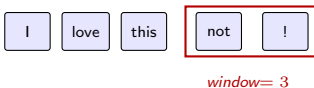
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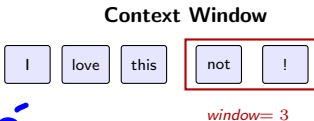
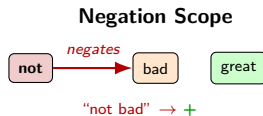
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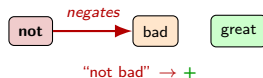
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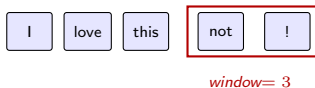
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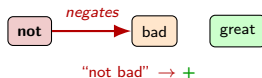
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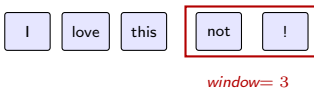
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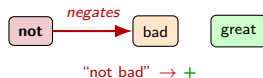
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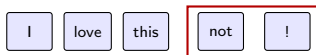
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Pre-processing
Curated

Eng

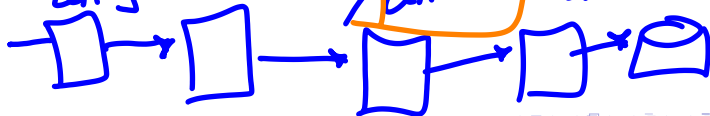
spa

LLM

SLM

ing. < raw spa.
raw org.

LLM gram LLM trans.



Anali.

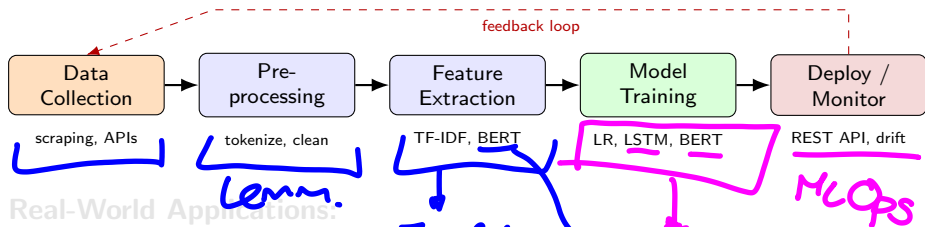
- ADP Google
- Strands

AI Agents

Task



End-to-End NLP Pipeline



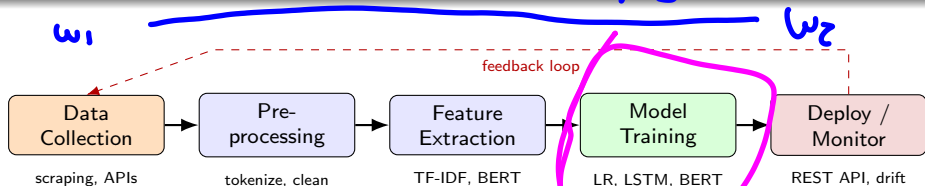
Real-World Applications:

- Chatbots & virtual assistants
- Machine translation (DeepL)
- Search engines
- Sentiment analysis
- Spam / phishing detection
- Code generation (Copilot)



End-to-End NLP Pipeline

500 pages



Real-World Applications:

- Chatbots & virtual assistants
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- ADRES
- NER in clinical records
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 - Code generation (GitHub Copilot)

Transfer Learning

fine-tuning

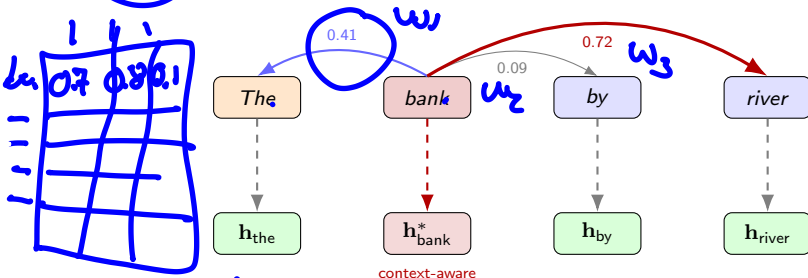


DL → 2012 ~
→ 2016 → He

Replace recurrence with **attention**: every token simultaneously queries **all** other tokens, capturing **long-range context** in parallel.

attention is all you need
river bank, not financial bank

"bank" attends most to "river" → river bank, not financial bank

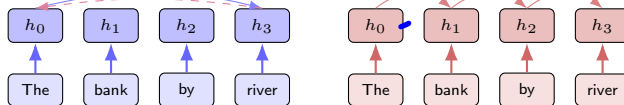


attention $\rightarrow$ linear algebra



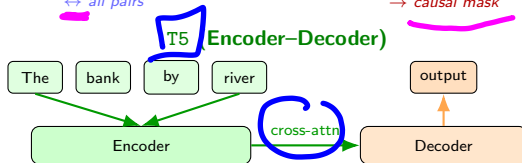
Transformer Families: Architecture Comparison

gpt ← BERT (Encoder) *→ Hugging Faces*



$\leftrightarrow$ all pairs

$\rightarrow$ causal mask



understand & generate: translation, summarisation

BERT pre-training

MLM: predict masked tokens

"The [MASK] by the river"

GPT pre-training

CLM: predict next token

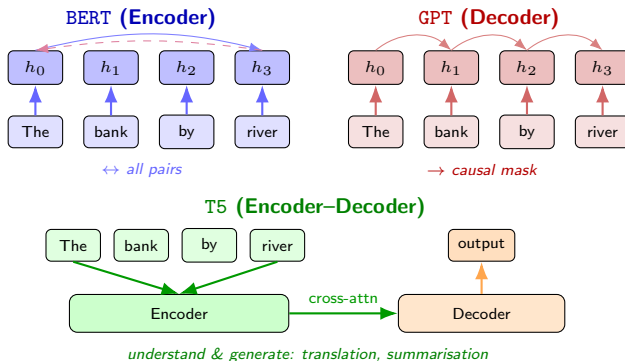
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T5 pre-training

Span corruption: mask spans

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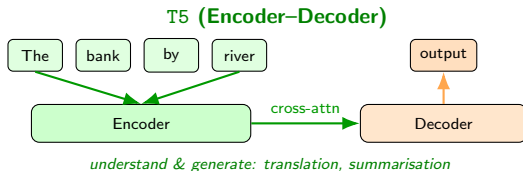
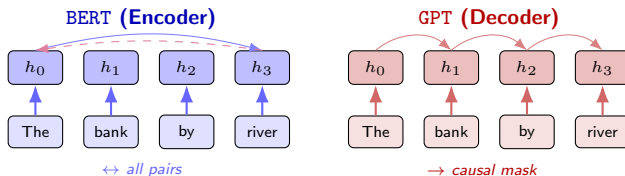
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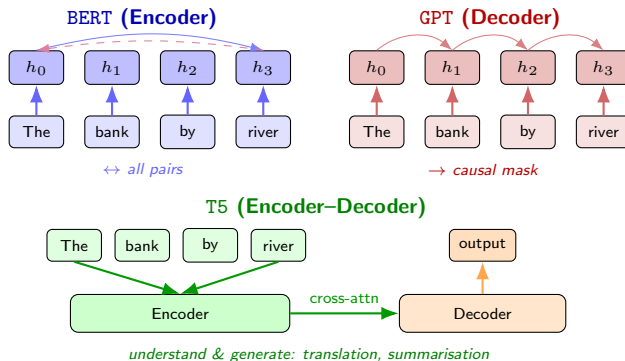
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mask spans

"The <X> by the river"

Handwritten notes in pink and blue:

- Top k* (pink)
- Top P* (pink)
- Word / P₁* (blue)
- word2 P₂* (blue)
- word3 P₃* (blue)

Transformer Families: Architecture Comparison



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GPT pre-training

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Sentence Embeddings & Semantic Search

Encoding a Sentence

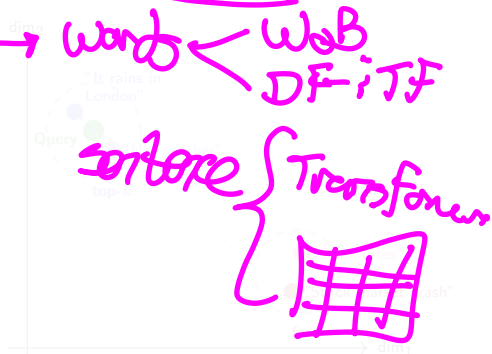
- Feed full sentence through BERT
- Pool [CLS] token or mean-pool hidden states
- Output: $\mathbf{v} \in \mathbb{R}^{768}$ — a semantic fingerprint

Cosine Similarity

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

1 $\approx$ same meaning

0 $\approx$ unrelated



Semantic search in embedding space

This is the Embed + Vector DB step in RAG



Sentence Embeddings & Semantic Search

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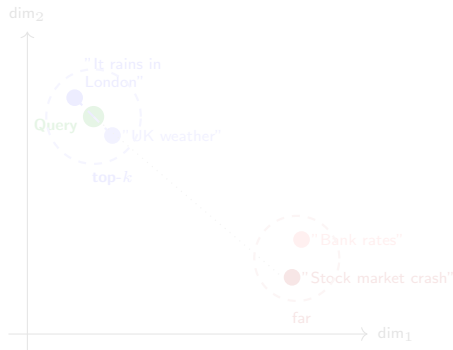
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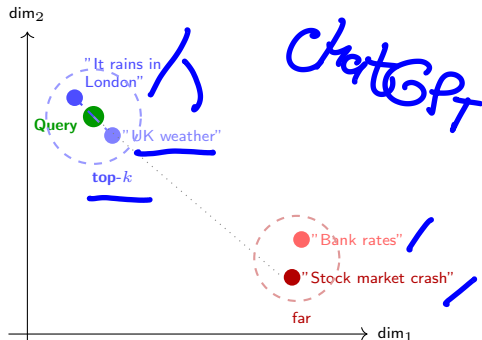
- Feed full sentence through BERT
- Pool [CLS] token or mean-pool hidden states
- Output: $\mathbf{v} \in \mathbb{R}^{768}$ — a **semantic fingerprint**

Cosine Similarity

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

1 $\approx$ same meaning

0 $\approx$ unrelated



Semantic search in embedding space

This is the Embed + Vector DB step in RAG

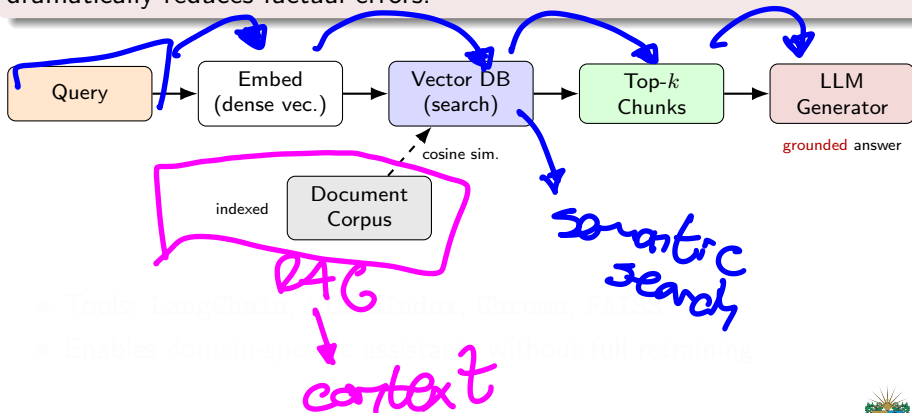
similarity DF-15F



Retrieval-Augmented Generation (RAG)

Motivation

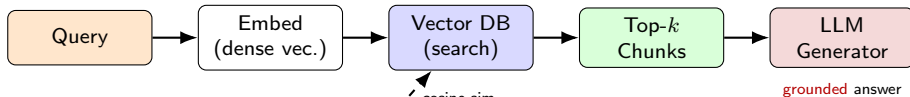
LLMs hallucinate. **Grounding** responses in a **verified document store** dramatically reduces factual errors.



Retrieval-Augmented Generation (RAG)

Motivation

LLMs hallucinate. **Grounding** responses in a **verified document store** dramatically reduces factual errors.



- real world*
- change at any time*
- Tools: LangChain, LlamaIndex, Chroma, FAISS
 - Enables **domain-specific** assistants without full retraining
- chatbot*



Current Challenges in NLP

- **Hallucination:** LLMs generate plausible but false content

- **Bias:** stereotypes inherited from training corpora

- **Low-resource languages:** ~7,000 languages; most resources are English

- **Interpretability:** SHAP, LIME, Attention maps as post-hoc explanations

- **Computational cost:** 100M to train; LoRA/PEFT for efficient fine-tuning

- **Privacy:** text often contains PII — GDPR compliance required

RAG (CAG)

→ Parameters → top k
→ top 7
→ Temperature

Workshop Connection

Build a complete pipeline: scrape reviews → preprocess → train sentiment classifier → deploy as a Streamlit app with live inference.



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 - **Computational cost:** GPT-4 $\approx$ \$200M to train; LoRA/PEFT for efficient finetuning
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- Handwritten notes:*
- fine tuning ✓
 - RAG → vocabulary

Workshop Connection

Build a complete pipeline: *scrape reviews* → *preprocess* → *train sentiment classifier* → *deploy as a Streamlit app with live inference.*



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Build a complete pipeline: scrape *reviews* $\rightarrow$ *preprocess* $\rightarrow$ *train sentiment classifier* $\rightarrow$ *deploy* as a Streamlit app with live *inference*.



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billions

Workshop

Build a complete pipeline: scrape news → preprocess → train sentiment classifier → deploy as a Streamlit app with live inference.



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SS

LoRA

3 weeks

~ 20000 GPUs

Workshop Course

Build a complete pipeline: scrape reviews $\rightarrow$ preprocess $\rightarrow$ train sentiment classifier $\rightarrow$ deploy as a Streamlit app with live inference.

Open source
Mislabel
Quen
Deepseek



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legal

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Outline

- 1 Foundations and Basic Concepts
- 2 Web Scraping and Data Collection
- 3 Sentiment Analysis
- 4 Applications and Integration



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/data-analysis-programming>

